

[Start Assignment](#)

- Due Jul 26 by 8:59pm
- Points 100
- Submitting a text entry box or a file upload
- Available Jun 28 at 9pm - Jul 29 at 8:59pm

Milestone #2

11 — Extension to Relational Database Management Systems



Due Sunday by 11:59 p.m. ET

Overview

In this module, you will analyze your case study by completing the second Milestone of your project: **Relational database implementation in MySQL and application to access database (Python or R)**. This Milestone is the second step in completing your Use Case Project for this course. For this milestone you will create your own database in MySQL and build a small analytical application using a Python or R language that connects to it.

You will not receive a grade for this assignment at this time, but you will receive feedback on your work. Use this feedback to refine your project and improve your work before submitting the final presentation at the end of this course.

Key Tasks

Milestone 2 | Relational database implementation in MySQL and Application written in Python or R to access your MySQL database

In Milestone #2, you are expected to build a physical MySQL database on your own machine, populate it with sample data, and run SQL queries against it. You will also develop a Python or R application that connects to your MySQL database, retrieves data, and performs basic analytics such as bar charts, histograms, pie charts, line charts, or scatter plots. Below are the expected steps. Please make sure you do all of the following:

Step 1| Create a database in MySQL for your project.

- Write the full SQL script to create the database and all required tables (DDL).
- Run the script and make sure the database and tables are created successfully.
- Take a screenshot showing the successful creation of the database and tables (e.g., from MySQL Workbench or terminal).

STEP 2 | Populate your tables with sample data.

- Provide the INSERT scripts you used, OR show clearly how data was generated/loaded using the tools provided by the professor.
- Your tables should have enough rows (around 20 rows in each table) to make the queries and plots meaningful.

STEP 3 | Write at least 5 - 8 SQL queries on your database to show that your database is working perfectly.

- Start from easier queries and move to more complex ones.
- At least some of these queries must involve multiple tables (JOINS).
- For each query, keep the query text and show the result (screenshot or copied output).

STEP 4| Create an application using a Jupyter notebook (.ipynb) or R Markdown (.Rmd).

- The notebook/Rmd must successfully connect to your MySQL database (For eg, you can use mysql connector in python to connect to your database. Make sure to provide your username, password and other details for the successful connection).
- In the notebook/Rmd, run at least 3–4 of your SQL queries from the database connection.
- Load the results into data frames (e.g., pandas/R data frames).
- From these results, create a minimum of 3 plots (e.g., bar charts, histograms, line plots, pie charts, etc.).
- Each plot should be based on data from your queries and should be readable and labeled.

SUBMISSION | Your submission should include following:

- The SQL script containing:
- CREATE DATABASE, CREATE TABLE statements
- Screenshot of successful database creation
- Data Insertion
- 5-8 Queries in the mySQL workbench
- Entire code (Jupyter notebook or R Markdown file):

What to Submit

- **File Type:** A Word or PDF Document
- **Length:** 8-10 pages (excluding citation/reference page)
- **Font:** 12-point Arial (or other sans serif)
- **Spacing:** One-inch margins on all sides, and double-spaced text
- **Citation:** APA Style. Support your arguments with examples, evidence, and references from the assigned readings as well as your own background research, where applicable. Any sources used should be cited. Learners are expected to independently develop an original submission.
- **Academic Integrity:** Learners are expected to independently develop an original submission. Copying from the Sober case in the textbook, from peers, or from online sources is not permitted. Any form of plagiarism or academic dishonesty will result in zero credit. Learners should use Turnitin to verify the originality of their work.
- **AI Use:** You are welcome to use AI tools as part of your work, but you must explain how they were used and clearly highlight the sections generated by AI. The rest of the work should reflect your own understanding and effort. You are also responsible for ensuring that all submission, including AI-generated material is used appropriately, is accurate, and complies with Northeastern University's Academic Integrity and Use of AI policy, which is linked in the syllabus.

11- Milestone #2 Rubric

Criteria	Ratings				Pts
MySQL Database Creation	20 to >18.0 pts Exceeds Standard Full DDL script is correct and runs without errors; all tables, keys, and constraints are properly created; screenshot clearly shows successful creation.	18 to >16.0 pts Standard Database and tables are created correctly but missing some minor details or refinements; screenshot included.	16 to >14.0 pts Approaching Standard Tables or keys are incomplete or partially incorrect; errors or missing elements in the DDL.	14 to >0 pts Below Standard Database creation is incorrect, unclear, or missing.	20 pts
Data Population	20 to >18.0 pts Exceeds Standard Tables are populated with sufficient and valid sample data (around 20 rows per table), with clear INSERT scripts or documented loading steps.	18 to >16.0 pts Standard Data is populated correctly but with slight gaps, fewer rows, or missing minor documentation.	16 to >14.0 pts Approaching Standard Limited or unclear data population; missing scripts or insufficient data for meaningful queries.	14 to >0 pts Below Standard Data population is unclear, very limited, or missing.	20 pts
SQL Queries	20 to >18.0 pts Exceeds Standard Includes 5–8 correct queries with increasing complexity; multiple JOINS; results shown clearly for each query.	18 to >16.0 pts Standard Queries run correctly but missing some complexity or detail; minor gaps in screenshots or output.	16 to >14.0 pts Approaching Standard Queries are partially correct; limited variety; missing outputs or JOINS.	14 to >0 pts Below Standard Queries are incorrect, unclear, too few, or missing.	20 pts
Application (Python or R)	20 to >18.0 pts Exceeds Standard Jupyter Notebook/Rmd connects successfully; runs 3–4 queries; loads results into data frames; includes at least 3 clear, labeled plots.	18 to >16.0 pts Standard Application connects and runs queries but missing some minor details or polish in plots or documentation.	16 to >14.0 pts Approaching Standard Connection works inconsistently; few queries run; plots incomplete or unclear.	14 to >0 pts Below Standard Application does not connect, run queries, or produce usable plots.	20 pts
Submission Quality	20 to >18.0 pts Exceeds Standard Submission is well organized, clearly written, and meets the page length expectation with high readability.	18 to >16.0 pts Standard Submission is clear and organized but has minor formatting or completeness gaps.	16 to >14.0 pts Approaching Standard Submission lacks clarity, organization, or does not fully meet length expectations.	14 to >0 pts Below Standard Submission is disorganized, unclear, or significantly incomplete.	20 pts
Total Points: 100					